

# A380X weather radar GAIN, ELEVN and TILT

The manual GAIN, ELEVN and TILT settings of the MFD SURV CONTROLS page reach the ND messages and the VD white line, and start from their FCOM default values

|                         |                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aircraft</b>         | A380X                                                                                                                                                                                         |
| <b>Where to find it</b> | MFD SURV > CONTROLS (WXR block: GAIN, ELEVN/TILT); ND, right side above the TERR box; VD                                                                                                      |
| <b>Branch</b>           | core/a380x-wxr-controls , based on core/nd-weather-terrain with core/a380x-mfd-fms merged in (stacked: needs core/nd-weather-terrain, core/surveillance and core/a380x-mfd-fms merged first). |
| <b>Status</b>           | ND messages tested in the sim by the user: OK (2026-10-02); FCOM defaults user-checked in the sim (2026-10-02); VD white line tested in the sim by the user: OK (2026-10-03).                 |
| <b>Pictures</b>         | No pictures: the messages and the line are drawn by the stock ND / VD code (no screenshot of this state).                                                                                     |

## 1. ND messages of the manual GAIN, ELEVN and TILT

**Aircraft:** A380X | **Where:** ND, right side above the TERR box | **Status:** Tested in the sim by the user: OK (2026-10-02).

### What it is for

The GAIN button and the ELEVN/TILT option list of the MFD SURV CONTROLS page kept their state but showed nothing. The ND now shows the manual radar settings as the A380 FCOM WXR MESSAGES (DSC-34-20-30-20 P 17).

### What you see

| SURV CONTROLS setting                                | ND message                                                                                                            |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| GAIN AUTO, ELEVN/TILT AUTO                           | <b>WX</b> in green (as before)                                                                                        |
| GAIN MAN, value 53                                   | <b>GAIN 53%</b> in cyan, above the mode line; with ELEVN/TILT at AUTO the mode line is empty and GAIN takes its place |
| ELEVN/TILT at TILT, value -12.0                      | <b>TILT -12.0°</b> in cyan instead of WX                                                                              |
| ELEVN/TILT at ELEVN, value 12500, QNH reference      | <b>ELEVN 12500FT</b> in cyan instead of WX                                                                            |
| ELEVN/TILT at ELEVN, STD baro reference on that side | <b>ELEVN FL125</b> in cyan                                                                                            |
| Manual mode without a value                          | GAIN ---%, TILT ---.°, ELEVN -----FT (or ELEVN FL---)                                                                 |
| MODE MAP                                             | MAP in green (GAIN nn% above it with GAIN MAN); no ELEVN / TILT message                                               |
| WXR OFF                                              | WXR OFF in white, no GAIN line                                                                                        |

### Limitations and why

| Limitation                                                  | Why                                                                                                                                       | What to do                                                 |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| The radar picture does not change with GAIN, ELEVN or TILT. | The MSFS radar has no gain and no height data; tilting its beam crashed the gauge.                                                        | The settings show in the messages and on the VD line only. |
| Cyan is a choice.                                           | The A380 FCOM gives no colour; cyan is the colour of the manual modes of the same radar family in the A320 FCOM (DSC-34-SURV-30-30 P 10). | -                                                          |

The grey background the FCOM figure prints behind GAIN is not drawn.

Not reproduced.

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## How to use

1. MFD **SURV** > **CONTROLS**, WXR block (WXR AUTO, MODE WX).
2. GAIN: press the button to **MAN**, enter a value in the field below it (0 to 100 %).
3. ELEVN/TILT: select **ELEVN** (enter an altitude or FL) or **TILT** (enter -15.0 to +15.0 degrees) in the field below the list.
4. Read the message on the ND (WX overlay selected). Back to **AUTO** for WX.

## How to test

| # | Do this                                                                                                        | Expected result                                         |
|---|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1 | A380X, CPT EFIS <b>WX</b> overlay, ND ARC. MFD SURV > CONTROLS: WXR AUTO, MODE WX, GAIN AUTO, ELEVN/TILT AUTO. | ND: <b>WX</b> in green on the right.                    |
| 2 | GAIN to <b>MAN</b> , enter <b>53</b> .                                                                         | ND: <b>GAIN 53%</b> in cyan, the WX message gone.       |
| 3 | ELEVN/TILT: select <b>TILT</b> , enter <b>-12</b> .                                                            | ND: <b>TILT -12.0°</b> in cyan, with GAIN 53% above it. |
| 4 | Select <b>ELEVN</b> , enter <b>12500</b> (QNH set on the CPT side).                                            | ND: <b>ELEVN 12500FT</b> in cyan.                       |
| 5 | Set the CPT baro reference to <b>STD</b> .                                                                     | ND: <b>ELEVN FL125</b> .                                |
| 6 | GAIN and ELEVN/TILT back to <b>AUTO</b> .                                                                      | ND: <b>WX</b> in green, no GAIN line.                   |
| 7 | <b>MODE MAP</b> with GAIN MAN.                                                                                 | ND: <b>MAP</b> in green with the GAIN line above it.    |
| 8 | WXR <b>OFF</b> (confirm).                                                                                      | ND: <b>WXR OFF</b> in white, no GAIN line.              |

## 2. GAIN, TILT and ELEVN take the FCOM defaults at the first selection

**Aircraft:** A380X | **Where:** MFD SURV > CONTROLS, WXR block | **Status:** User-checked in the sim after a flight reload (2026-10-02): GAIN 50 %, TILT +3.0° on the ground, ELEVN 100 FT at CYUL.

## What it is for

A380 FCOM DSC-34-20-30-20-10 (GAIN and ELEVN knobs): "The default GAIN value is 50 %", "The default ELEVN value is the current aircraft altitude", the default TILT value is +3.00° on ground and 0° in flight. The page applies a default only when nothing is entered, but a value nobody had written read 0, a valid entry: after a power-up the first GAIN MAN, TILT or ELEVN selection kept 0. The MFD now starts with the three settings at no entry.

## What you should see

| Situation                           | What you should see                                                  |
|-------------------------------------|----------------------------------------------------------------------|
| First GAIN MAN after the MFD starts | GAIN field 50 %; ND GAIN 50%.                                        |
| First TILT selection on the ground  | TILT +3.0°; ND TILT +3.0°.                                           |
| First TILT selection in flight      | TILT +0.0°.                                                          |
| First ELEVN selection               | The current altitude, rounded to 100 ft (FL with the STD reference). |
| Later selections                    | The value entered or set before is kept.                             |

## How to use

1. Select GAIN **MAN**, **ELEVN** or **TILT** on the SURV CONTROLS page: the default value is entered.
2. Overwrite it in the entry field if needed.

## How to test

| # | Do this                                                                                       | Expected result                                                                     |
|---|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | A380X on the ground, after a flight load or reload. MFD SURV > CONTROLS: GAIN to <b>MAN</b> . | GAIN field <b>50 %</b> ; ND (WX overlay) <b>GAIN 50%</b> .                          |
| 2 | ELEVN/TILT: select <b>TILT</b> .                                                              | TILT field <b>+3.0°</b> ; ND <b>TILT +3.0°</b> .                                    |
| 3 | Select <b>ELEVN</b> .                                                                         | ELEVN field = the airport elevation rounded to 100 ft (for example 100 FT at CYUL). |
| 4 | Enter TILT -5, go to AUTO, then select TILT again.                                            | TILT <b>-5.0°</b> is kept (no default over an entry).                               |
| 5 | Reload the flight, take off, and select TILT for the first time in flight.                    | TILT <b>+0.0°</b> .                                                                 |

## 3. VD white line of the manual ELEVN and TILT

**Aircraft:** A380X | **Where:** VD under the ND | **Status:** Tested in the sim by the user: OK (2026-10-03).

### What it is for

A380 FCOM DSC-34-20-30-20: "The VD displays a white line indicating the elevation or tilt value". In manual ELEVN the VD shows a horizontal white line at the selected altitude; in manual TILT a straight white line from the aircraft symbol at the selected angle (zero tilt = the horizon as seen by the ADIRS), as in the FCOM VD figures.

### What you should see

| Situation                                           | What you should see                                                                                        |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| ELEVN selected                                      | Horizontal white line at the selected altitude, from the aircraft to the end of the VD range.              |
| TILT selected                                       | Straight white line from the nose of the aircraft symbol, up for a positive tilt, down for a negative one. |
| Selected altitude off the VD scale                  | No line.                                                                                                   |
| ELEVN/TILT AUTO, MODE MAP, WXR OFF or no WX overlay | No line (it goes with the ELEVN / TILT message of the ND).                                                 |

### Limitations and why

| Limitation                                                      | Why                                                                          | What to do |
|-----------------------------------------------------------------|------------------------------------------------------------------------------|------------|
| The line shows only while that side's ND shows the WX function. | Design choice: the line goes with the ND message.                            | -          |
| The tilt line is straight.                                      | Design choice, as in the FCOM figure: no earth curvature or beam refraction. | -          |
| The weather on the VD does not follow ELEVN or TILT.            | The MSFS radar has no height data.                                           | -          |

## How to use

1. Select the **WX** overlay on the EFIS panel and ND ARC or ROSE NAV (the pages with the VD).
2. On MFD SURV > CONTROLS select **ELEVN** or **TILT** and enter the value.
3. Read the white line on the VD.

## How to test

| # | Do this                                                                                                                                     | Expected result                                                                                 |
|---|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1 | A380X in flight, for example at 10 000 ft. CPT EFIS WX overlay, ND ARC 40 NM. SURV CONTROLS: ELEVN/TILT <b>ELEVN</b> , enter <b>12000</b> . | VD: a horizontal white line at 12 000 ft, ahead of the aircraft symbol.                         |
| 2 | Select <b>TILT</b> , enter <b>+3</b> .                                                                                                      | VD: a straight white line from the nose of the aircraft symbol, rising to the end of the range. |
| 3 | Enter <b>-3</b> .                                                                                                                           | The line goes down from the aircraft symbol.                                                    |
| 4 | Select <b>ELEVN</b> with a value far above the VD scale (for example 60000).                                                                | No line.                                                                                        |
| 5 | ELEVN/TILT back to <b>AUTO</b> .                                                                                                            | The line goes; ND shows WX.                                                                     |
| 6 | With TILT selected, deselect the WX overlay.                                                                                                | The line goes with the ND message.                                                              |